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Studies of muons in extensive air showers from ultra-high energy cosmic rays observed with the Telescope Array surface detector

R. Takeishi[1], H. Sagawa[1], M. Fukushima[1], M. Takeda[1], T. Nonaka[1], K. Kawata[1], E. Kido[1], N. Sakurai[2], T. Okuda[3], S. Ogio[2], J. N. Matthews[4], B. Stokes[4] for the Telescope Array Collaboration

[1]*Institute for Cosmic Ray Research, University of Tokyo, Kashiwa City, Chiba, Japan*
[2]*Graduate School of Science, Osaka City University, Osaka City, Osaka, Japan*
[3]*Graduate School of Science and Engineering, Ritsumeikan University, Kyoto City, Kyoto, Japan*
[4]*Department of Physics and Astronomy, University of Utah, Salt Lake City, Utah, USA*

E-mail: take@icrr.u-tokyo.ac.jp

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The number of muons in the air shower induced by ultra-high energy cosmic rays (UHECRs) has been measured with surface detector (SD) arrays of various experiments. Monte Carlo (MC) prediction of the number of muons in air showers depends on hadronic interaction models and the primary cosmic ray composition. By comparing the measured number of muons with the MC prediction, hadronic interaction models can be tested. The Pierre Auger Observatory reported that the number of muons measured by water Cherenkov type SD is about 1.8 times larger than the MC prediction for proton with QGSJET Ⅱ-03 model. The number of muons in the Auger data is also larger than the MC prediction for iron. The Telescope Array experiment adopts plastic scintillator type SD, which is sensitive to the electromagnetic component that is the major part of secondary particles in the air shower. To search for the high muon purity condition in air showers observed by the TA, we divided air shower events into subsets by the zenith angle θ, the azimuth angle ϕ relative to the shower arrival direction projected onto the ground, and the distance R from shower axis. As a result, we found subsets with the high muon purity ~65%, and compared the charge density between observed data and MC. The typical ratios of the charge density of the data to that of the MC are 1.71 ± 0.10 at 1870 m $< R <$ 2150 m and 3.24 ± 0.40 at 2850 m $< R <$ 3280 m. The difference in the charge density between the data and the MC is larger at the higher muon purity. These results imply that the excess of the charge density in the data is partly explained by the muon excess.

KEYWORDS: muon, ultra-high energy cosmic ray, hadronic interaction

1. Introduction

Cosmic rays with energies above about 10^{18} eV are called ultra-high energy cosmic rays (UHECRs). The origin of UHECR is a long-standing mystery. When UHECRs enter in the atmosphere, they interact with atmospheric nuclei and generate the particle cascade, which is called air shower. The Telescope Array (TA) experiment [1] is designed to measure air shower secondary particles on the ground with the surface detector (SD) and fluorescence light induced from the air shower with the fluorescence detector (FD). The TA experiment aims to reveal the origin of UHECR by measuring the spectrum, the anisotropy and the mass composition of UHECR. UHECRs are observed by the signal of air shower particles, and information of primary cosmic rays is estimated from the air shower Monte Carlo (MC) simulation. UHECR air showers are not fully understood, and the observation results of the mass composition have the dependence of hadronic models used in the air

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shower MC [6, 7]. Recently, the Pierre Auger Observatory reported [2] that the average ratio of observed number of muons to MC prediction is $1.841 \pm 0.029(\text{stat.}) \pm 0.324(\text{syst.})$ using QGSJET II-03 model [3] for proton. The number of muons in the Auger data is also larger than the MC prediction for iron. Thus, the Auger group claimed the excess of muons in the data.

The number of muons from the UHECR on the ground depends on the composition of primary cosmic rays. Its MC prediction also depends on hadronic interaction models. By comparing the measured number of muons with the MC prediction, hadronic interaction models can be tested. The TA experiment uses the SD made of plastic scintillator [4]. It is sensitive to the electromagnetic (EM) component that is the major part of secondary particles from the air showers. An analysis approach to increasing muon purity is necessary to study the number of muons in the TA SD data. This work aims to develop an analysis of muons from the UHECR air showers by using TA SD and study the difference of the number of muons between the data and the MC.

The analysis strategy is as follows:

(1) Assume the cosmic ray mass composition is proton.

(2) Search for the analysis condition where the muon purity in the SD signal becomes high using the MC.

(3) Compare the observed charge density from air shower particles with the MC prediction on the high muon purity condition.

2. Method

The TA SD array is located in western Utah, USA and consists of 507 scintillation counters, placed on a square grid with 1.2 km spacing, covering 700 km^2. The three TA FDs are located around the SD array, looking towards the sky above the array.

We use the TA SD 7 years' dataset recorded from 11 May 2008 through 11 May 2015, and the events reconstructed by the same method as the TA spectrum analysis [5] with energy range $10^{18.8}$ eV $< E < 10^{19.2}$ eV. In the energy range, the composition of the primary cosmic rays is consistent with proton within statistical and systematic errors by X_{max} measurement using the TA FD [6–9], thus we use the MC for proton primaries. The experimental data is compared with the MC using the hadronic models QGSJET II-03, QGSJET II-04 [10], Epos 1.99 [11] and Sibyll 2.1 [12].

The EM components are composed of electron and gamma. They are attenuated faster than muons compared in the same pass length, because the EM components largely lose their energy by the pair production and bremsstrahlung in the shower propagation but muons can penetrate the atmosphere down to the ground before they decay. Hence the ratio of the number of air shower muons to that of all charged particles in the same SDs (hereafter this ratio is described as the muon purity) is expected to be larger as the SDs are more distant from secondary particle generation points on the shower axis. We divide the air shower events of the dataset using θ (the zenith angle), ϕ (the azimuth angle relative to the shower arrival direction projected onto the ground), and R (the distance from shower axis). The geometry definition of ϕ is described in figure 1. When θ, $|\phi|$ or R values become large, the pass length of air shower particles increases, then the muon purity in the signal of SDs is expected to be high.

For each event, SDs which have air shower signals are picked up and the integrated FADC for each waveform is calculated. The value by the unit of VEM (Vertical Equivalent Muon) is filled in the histogram of corresponding (θ, ϕ, R) bin. Figure 2 shows the sample waveforms and histograms in each particle type. One detector signal deserves one entry in the histogram. If there is a SD in a (ϕ, R) bin but it does not have any signal, then one statistics is filled to the corresponding bin of the histogram as 0 VEM. Figure 3 shows the lateral distributions of the charge density and the muon purity. We calculate the muon purity as the ratio of the average charge density of the muons to that of the total component. For the condition ($30^\circ < \theta < 45^\circ$, $150^\circ < |\phi| < 180^\circ$, 2000 m $< R <$ 4000 m), the muon purity is 60 - 70%, and we use this condition for the comparison of the charge density of

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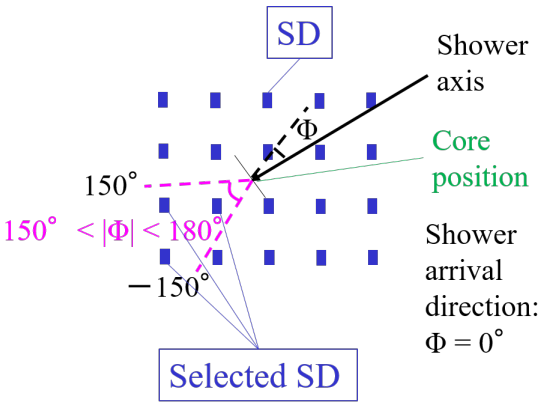


Fig. 1. ϕ definition. SD location on the ground is selected by ϕ to reduce EM background. The muon purity in the SD signal is calculated in each ϕ bin. There are six regions for the analysis and dashed magenta lines show boundaries of $150° < |\phi| < 180°$.

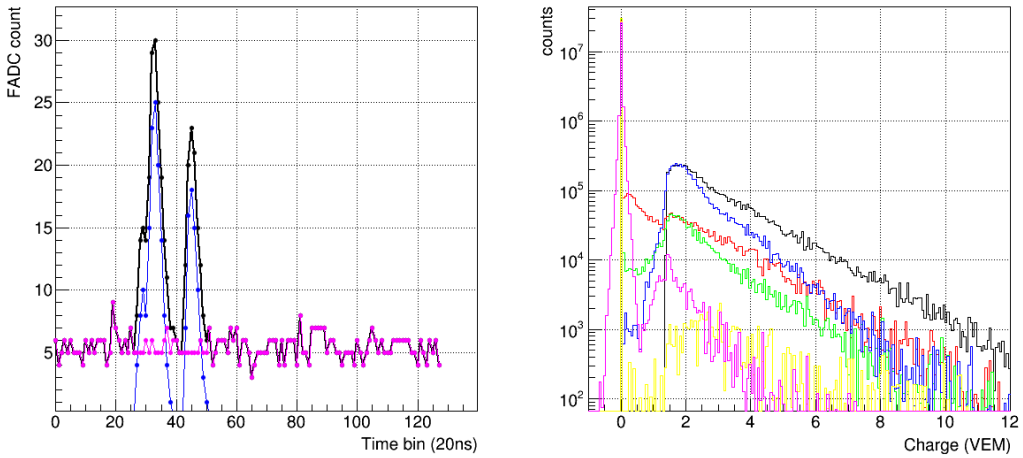


Fig. 2. Sample event waveforms and charge distribution in a SD (3 m^2 in area) for different particle types by the MC. The red, green, blue, yellow, magenta and black represent gamma, electron, muon, other shower components, atmospheric muon background and the total of them, respectively. (left) Sample event waveforms in a SD for $E = 5.67$ EeV, $\theta = 38.6°, |\phi| = 175°, R = 1980$ m. (right) Histograms of the charge density in a SD for $10^{18.8}\text{eV} < E < 10^{19.2}\text{eV}, 30° < \theta < 45°, 150° < |\phi| < 180°, 1870\text{ m} < R < 2150$ m.

the data with that of the MC.

3. Results

Figure 4 shows the lateral distributions of the charge density, and the ratio of the data to the MCs with various hadronic interaction models; QGSJET Ⅱ-03, QGSJET Ⅱ-04, Epos 1.99 and Sibyll 2.1. The observed lateral distribution (black points) falls down slower than the MCs with all the hadronic interaction models (colors).

The typical ratios of the charge density between the data and the MC are calculated to be 1.71 ± 0.10 at $1870\text{ m} < R < 2150$ m and 3.24 ± 0.40 at $2850\text{ m} < R < 3280$ m. Figure 5 shows the correlation

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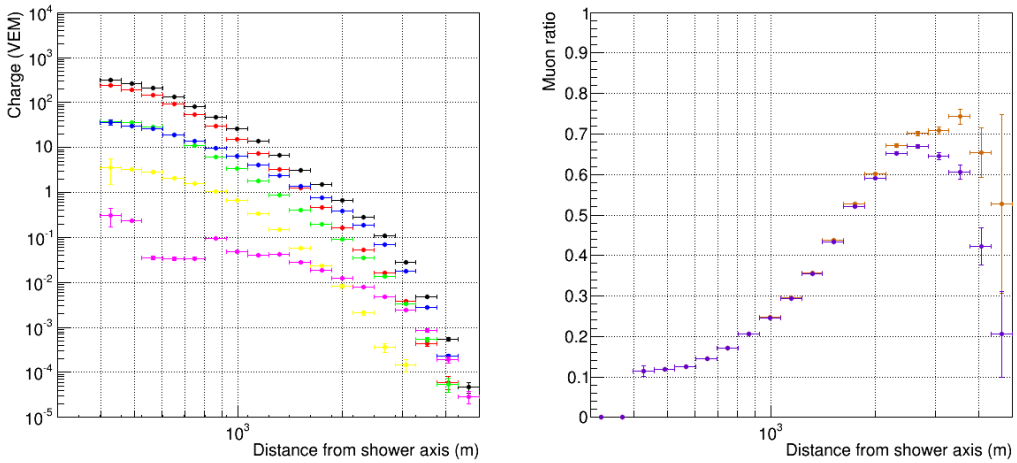


Fig. 3. Lateral distributions of air showers of the MC with QGSJET II-03 model for $30° < \theta < 45°, 150° < |\phi| < 180°$ and 500 m $< R <$ 5000 m. The horizontal axis describes R, which is the distance from shower axis. The vertical error bar shows the standard deviation. (left) Lateral distributions of the average charge density. The red, green, blue, yellow, magenta and black represent gamma, electron, muon, other shower components, atmospheric muon background and the total of them, respectively. (right) Lateral distributions of the muon purity. The violet and orange show the calculation with and without atmospheric muon background, respectively.

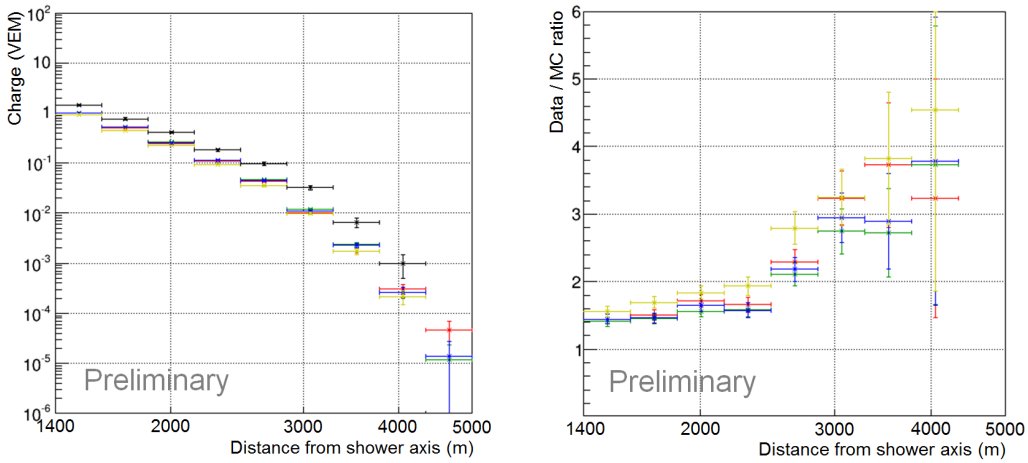


Fig. 4. Lateral distributions of air showers of the data and the MC with various hadronic models for $30° < \theta < 45°, 150° < |\phi| < 180°$ and 1400 m $< R <$ 5000 m. (left) Lateral distributions of the average charge density assuming the histograms follow the Poisson distribution. The black, red, blue, green and yellow represent data, QGSJET II-03, QGSJET II-04, Epos 1.99 and Sibyll 2.1, respectively. (right) Average ratio of the data to the MC. The color corresponds to MC hadronic models described in the left figure.

between the muon purity expected from the MC and the ratio of the charge density of the data to that of the MC. The muon purity and the ratio of the data to the MC on the high muon purity condition ($30° < \theta < 45°, 150° < |\phi| < 180°$, 2000 m $< R <$ 4000 m, magenta cross in figure 5) are 65% and 1.88 ± 0.08, respectively. In the case of the low muon purity condition ($\theta < 30°, |\phi| < 30°$, 2000 m $< R <$ 4000 m, black open circle in figure 5), the muon purity and the ratio of the data to the MC are

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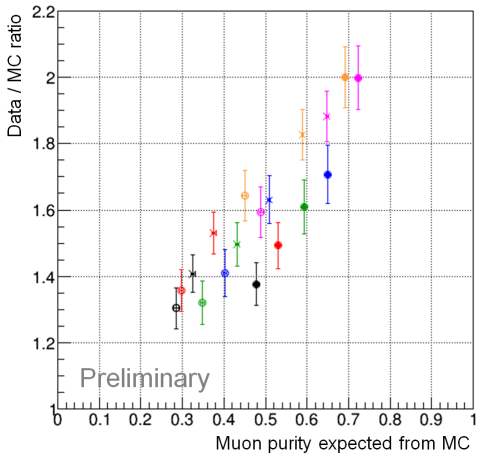


Fig. 5. The correlation between the muon purity and the ratio of the charge density of the data to that of the MC with QGSJET II-03 model for 2000 m $< R <$ 4000 m. The black, red, green, blue, orange and magenta represent $|\phi| < 30°$, $30° < |\phi| < 60°$, $60° < |\phi| < 90°$, $90° < |\phi| < 120°$, $120° < |\phi| < 150°$ and $150° < |\phi| < 180°$, respectively. The open circle, cross and filled circle represent $\theta < 30°$, $30° < \theta < 45°$ and $45° < \theta < 55°$, respectively.

calculated to be 28% and 1.30 ± 0.06, respectively. This figure shows larger difference in the charge density between the data and the MC at the higher muon purity. The systematic uncertainty in this study is mainly caused by the uncertainty of the TA FD energy measurement 21% [13]. It causes the uncertainty of the charge density of the data, which is about 21%. The details of minor systematic uncertainties are under investigation. These results imply that part of the discrepancy in the charge density between the data and the MC is explained by the muon excess in the data.

4. Summary

We studied muons in the air shower induced by UHECR with the TA SD to test hadronic interaction models of the air shower MC. The air shower events and the locations of SDs were divided on θ, ϕ and R conditions to search for the high muon purity analysis condition. We compared the charge density from air shower signals of the data with that of the MC on the high muon purity condition ($E \simeq 10$ EeV, $30° < \theta < 45°$, $150° < |\phi| < 180°$, 2000 m $< R <$ 4000 m). On that condition, the muon purity expected from the MC is ~65%, and the typical ratios of the charge density of the data to that of the MC are 1.71 ± 0.10 at 1870 m $< R <$ 2150 m and 3.24 ± 0.40 at 2850 m $< R <$ 3280 m. The lateral distribution of the data falls down slower than the MC on the high muon purity condition, causing the larger ratio of the charge density of the data to that of the MC at the larger R value. The difference in the charge density between the data and the MC is larger at the higher muon purity, which implies that part of the discrepancy in the charge density between the data and the MC is explained by the muon excess in the data.

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